

NERVE IMPULSE

WHEN NERVE CELLS, OR NEURONS, ARE STIMULATED THEY UNDERGO CHEMICAL CHANGES THAT PRODUCE TINY WAVES OF ELECTRICITY – NERVE IMPULSES.

Information is conveyed throughout the nervous system as nerve impulses, or action potentials. Impulses are about 100 millivolts (0.1 volts) in strength and last just 1 millisecond ($1/1000$ s). The information carried depends on the location of the impulses in the nervous system, and on their frequency – from one impulse every few seconds to several hundreds per second.

When impulses reach a junction known as a synapse, they trigger the release of chemicals called neurotransmitters. Molecules of the neurotransmitter cross the synapse and stimulate the receiving neuron to fire an impulse of its own, as wave-like movements of ions (electrically charged particles). Neurotransmitters may also actively inhibit a receiving neuron from firing.

Presynaptic membrane

Membrane of sending cell's axon

Neurotransmitter

Molecule that flows across the synaptic cleft in about 1 millisecond, passing on the nerve impulse in chemical form

Synaptic vesicle

Package of neurotransmitter molecules that fuses with the cell membrane when an impulse arrives, releasing the molecules

Mitochondrion

Standard cellular component that provides energy

Postsynaptic membrane

Membrane of receiving cell's dendrite

Neurotubule

Specialized microtubule that conveys synaptic vesicles from the cell body to the axon terminal

Synaptic knob

Enlarged end of axon terminal

Membrane channel protein

Complex protein embedded in cell membrane; when enough ions flood through the channel, they cause a response in the receiving cell

Positive ion

Receptor

Site in membrane channel into which neurotransmitters slot, letting in charged ions

